

Kaan Bahtiyar

Ph.D. Candidate in Robotics | Precision Motion Control and Mechatronics
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SUMMARY

Ph.D. candidate in Robotics (expected June 2027) who designs, tunes and implements controllers that make precision machines settle faster and move more accurately: learned feedforward, auto-tuned feedback, active damping and disturbance compensation. Works from dynamic modeling and system identification through to real-time implementation, and validates every method on real hardware. First author of eleven papers, including three journal articles (*CIRP Annals*, *CIRP JMST*, *ASME JMSE*), co-inventor on a pending patent with an industrial partner, and first place among nine teams in the 2025 ASPE Student Challenge. Seeking roles in precision motion control and mechatronics.

EXPERIENCE

Graduate Research Assistant, Oregon State University

Corvallis, OR, 2020 – Present

Precision Manufacturing & Automation Laboratory

Precision motion control and mechatronics

- **Active inertial damper control:** Designed feedback and feedforward controllers that suppress residual vibration after rapid moves on a feed-drive testbed, each tuned from measured data within the stroke and force limits.
 - **Learned feedforward:** Cut the settling time to $\pm 2 \mu\text{m}$ by 98% (29.7 to 0.6 ms) and the peak-to-peak error by 90% (35 to 3.5 μm) versus feedback only, by using iterative learning control over about 10 runs.
 - **Auto-tuned feedback:** Formulated the tuning of direct velocity feedback control as a constrained quadratic program that selects the passband and gain across frequencies automatically: 70% less actuator stroke and up to 28% faster settling than conventional feedback control.
- **Cross-axis vibration compensation:** Designed a trajectory pre-filter that cancels Z tool-tip vibration excited by Y-axis motion on a 3-axis machining center, through the Z-axis feed drive, leaving the servo loops unchanged. Machine-in-the-loop tuning gives up to 77% lower vibration (5.5 to 1.3 μm).

Machining process control on a 3-axis CNC machine tool

- **Spindle-synchronous disturbance compensation:** Cancelled the effect of tool runout with a micro-circular trajectory, synchronized with the spindle, in the 5 kHz position loop of the X and Y feed drives, tuned online by convex optimization in 2 to 3 updates (10 to 15 spindle revolutions): 86% lower cutting-force variation (101 to 14 N). Patent pending.
- **Real-time spindle speed optimization:** Built a model-free extremum-seeking controller that optimizes the spindle speed from accelerometer feedback through the CNC's spindle-override input: 85% lower RMS vibration (131 to 20 m/s^2), and 57 to 66% lower than a conventional chatter mitigation strategy.
- **Real-time chatter detection:** Developed two lightweight detectors (spectral power ratio and recursive PCA) that separate chatter from forced vibration and flag its onset with a single threshold.

Mentoring

- Trained graduate and undergraduate students in system identification and controller design; guided two M.Sc. theses.

Graduate Teaching Assistant, Oregon State University

Corvallis, OR, 2020 – Present

- Led labs and review lectures for *Computer Control of Manufacturing Processes* (senior level), ran labs for *Precision Motion Generation* (graduate level), and contributed material for *Digital Control of Mechatronic Systems*.

R&D Undergraduate Senior Design Project, Arçelik

Istanbul, Turkey, 10/2018 – 6/2019

- Built parametric 3D CAD models of a washing-machine platform in Siemens NX, and linked NX, MATLAB and Excel into a fully automated particle-swarm optimization loop for weight and structural performance.

TECHNICAL SKILLS

- **Control design:** Feedback and feedforward motion control; iterative learning control; iterative feedback tuning; extremum-seeking and adaptive control; active damping and vibration suppression; trajectory generation and pre-filtering.
- **Modeling and identification:** First-principles and data-driven modeling of feed drives and machine structures; system identification; FRF-based modal analysis; machining process modeling and stability analysis.
- **Optimization and signal processing:** Convex and numerical optimization; constrained quadratic programming; least squares; regression; spectral analysis; principal component analysis; Kalman filtering.
- **Hardware:** Linear-motor XY stages; feed-drive and voice-coil stages; voice-coil and linear-motor inertial actuators; CNC machine tools; linear and rotary encoders, laser encoder, capacitive displacement sensors, accelerometers, dynamometers.
- **Real-time and software:** dSPACE and NI myRIO real-time control; CNC and PLC interfacing; MATLAB/Simulink, LabVIEW, Siemens NX.

EDUCATION

- Oregon State University** Corvallis, OR
– **Ph.D.**, Robotics 2022–Present
Expected June 2027. Advisor: Prof. Burak Sencer
- **M.Sc.**, Robotics 2020–2023
Thesis: *Detection of Chatter Vibrations and Adaptive Spindle Speed Optimization in Milling Process*
- **Relevant Coursework:** Kinematics Dynamics & Control · Mechanical Vibrations · Learning Based Control · Estimation Filtering & Detection · Convex Optimization · Optimization for AI
- Istanbul Technical University** Istanbul, Turkey
– **B.Sc.**, Control and Automation Engineering 2017–2020
– **B.Sc.**, Mechanical Engineering 2014–2019
– **Relevant Coursework:** Feedback Control Systems · Control System Design · Dynamic System Modeling & Simulation · Intelligent Control Systems · Servo Motors · Strength of Materials · Manufacturing Processes

HONORS

- ASPE Student Challenge – 1st Place Winner** (3-person team) 2025
– 1st place out of 9 competing teams at the 40th Annual Meeting of ASPE (San Diego, CA).
- Thomas A. Dow Student Scholarship** – American Society for Precision Engineering 2024
– Awarded at the 39th Annual Meeting (Houston, TX) for the paper “Data-Driven Feedforward Control of Active Inertial Dampers for High-Performance Vibration Mitigation.”
- Young Research Award** – Japan Society for Precision Engineering (ICPE2024) 2024
– Awarded at the 20th International Conference on Precision Engineering (Sendai, Japan) for the paper “Compensation of Axis-Coupled Inertial Forced Vibrations using Machine Tool Feed Drives.”
- NSF Travel Awards** – MSEC2024, NAMRC2022 2024, 2022
- Ranked 3rd in Department and Honor List** – Control & Automation Eng., Istanbul Technical University 2020

PATENTS & INTELLECTUAL PROPERTY

Control Device, Machine Tool System, and Method For Controlling Machine Tool.

Inventors: Burak Sencer, Kaan Bahtiyar, Levent Bozcu. Patent application pending (2025).

PROFESSIONAL SERVICE

Journal reviewer, *ASME Journal of Manufacturing Science and Engineering* (2 manuscripts reviewed).
Journal reviewer, *Measurement*, Elsevier, the journal of IMEKO (1 manuscript reviewed).

PUBLICATIONS

First author of all eleven publications below.

Journal Publications

Kaan Bahtiyar, Burak Sencer, Ryosuke Ikeda. “Data-driven Adaptive Compensation of Tool Runout in Milling Via Machine Tool Feed Drives.” *ASME Journal of Manufacturing Science and Engineering*. 2026.

Kaan Bahtiyar, Burak Sencer. “Model-free Optimal Spindle Speed Control for Real-Time Vibration Mitigation in Milling.” *CIRP Journal of Manufacturing Science and Technology*. 2025.

Kaan Bahtiyar, Burak Sencer, Xavier Beudaert. “Data-Driven Feedforward Control of Inertial Dampers for Accuracy Improvement.” *CIRP Annals*. 2024.

Conference Publications

Peer-Reviewed Full Papers (Published in Proceedings)

Kaan Bahtiyar and Burak Sencer. “Adaptive spindle speed control for real-time machining vibration mitigation.” *11th International Conference on Leading Edge Manufacturing in 21st Century (LEM21)*. 2025.

Kaan Bahtiyar, Levent Bozcu, Seyed Mahmood Shantiaeezade, Burak Sencer, Ryosuke Ikeda, Norikazu Suzuki. “Adaptive Tool Eccentricity Compensation using Machine Tool Feed Drives.” *MM Science Journal*, Special Issue on HSM 2025, 2025.

Kaan Bahtiyar and Burak Sencer. “Brief paper: Model-free adaptive spindle speed regulation for mitigation of machining vibrations.” *ASME International Manufacturing Science and Engineering Conference (MSEC 2024)*. 2024.

Kaan Bahtiyar and Burak Sencer. “Lightweight and Robust Chatter Detection Algorithms for Milling.” *Manufacturing Letters* 33, Special Issue on NAMRC50, 2022.

Abstract-Reviewed Extended Abstracts (Published in Proceedings)

Kaan Bahtiyar, Burak Sencer, and Xavier Beudaert. “Data-driven feedforward control of active inertial dampers for high-performance vibration mitigation.” *39th Annual Meeting of the American Society for Precision Engineering*. 2024. (Oral presentation.)

Kaan Bahtiyar and Burak Sencer. “Extremum seeking control based real-time optimization of spindle speed in machining.” *37th Annual Meeting of the American Society for Precision Engineering*. 2022. (Poster presentation.)

Kaan Bahtiyar and Burak Sencer. “Time-domain based early detection of chatter vibrations.” *36th Annual Meeting of the American Society for Precision Engineering*. 2021. (Poster presentation.)

Peer-Reviewed Full Papers (No Proceedings)

Kaan Bahtiyar, Eiji Shamoto, and Burak Sencer. “Compensation of axis-coupled inertial forced vibrations using machine tool feed drives.” *20th International Conference on Precision Engineering (ICPE 2024)*. 2024.